# The Fitzpatrick Extension Need Not Be Quasidense
# A Later-Literature Resolution of arXiv:1612.02500

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**Status.** Already answered in the literature. Simons's later arXiv:2407.00479 gives an explicit negative answer, so this target is not a new-result candidate.

## 1 Source question

Let $S : E \rightrightarrows E^*$ be closed, monotone, and quasidense. The source paper proves that its Fitzpatrick extension $S^{\mathrm{F}} : E^* \rightrightarrows E^{**}$ is maximally monotone, then asks whether $S^{\mathrm{F}}$ must itself be quasidense [1, Remark 3.10, p. 9].

$\partial \mathbb{I}_{K^*} = \partial \sigma_K$. The result now follows by using the technique of Lemma 3.5. $\sqcup$

**Remark 3.10.** [23, Theorem 12.4(a), p. 1047] implies that if $S\colon\ E \rightrightarrows E^*$ is closed, monotone and quasidense and $(y^*, y^{**}) \in E^* \times E^{**}$ then

$$(y^*, y^{**}) \in G(S^{\mathbb{F}}) \iff \inf_{(s,s^*)\in G(S)} \langle s^* - y^*, \widehat{s} - y^{**}\rangle = 0.$$

This is equivalent to the result proved in (9.3). There are more characterizations of $S^{\mathbb{F}}$ in Theorem 7.5 and (9.4). We do not know if $S^{\mathbb{F}}\colon E^* \rightrightarrows E^{**}$ is necessarily quasidense in the context of Theorem 3.6, but Theorems 3.7 and 2.14 show that *if* $f \in \mathcal{PCLSC}(E)$ *then* $(\partial f)^{\mathbb{F}}$ *is quasidense.* Theorem 3.7 is equivalent to [8, Théorème 3.1, pp. 376–378].

Lemma 3.3 is the "easy half" of Theorem 3.7. Theorem 3.7 and its two consequences, Theorems 3.8 and 3.9, will not be used any more in this paper, while Lemma 3.4 will be used in Theorem 8.3, and Lemma 3.5 will be used in Theorem 6.2. We have separated the proof of Theorem 3.7 into these two parts

Figure 1: The question in arXiv:1612.02500v4, PDF page 9.

## 2 Explicit later answer

The 2024 paper *Faces of quasidensity* uses $M^\sharp$ for the Gossez extension of a maximally monotone set $M \subset E \times E^*$. For $M = G(S)$ this is exactly the graph of the source paper's Fitzpatrick extension: both consist of the pairs $(y^*, y^{**})$ monotonically related to the canonical image of $G(S)$. For maximally monotone sets, quasidensity is equivalent to type (NI) [2, Theorem 11.1].

Define

$$U : \ell_1 \longrightarrow c_0, \qquad (Ux)_n = \sum_{k\ge n} x_k,$$

and let $S = U^{-1} : c_0 \rightrightarrows \ell_1$, so

$$G(S) = \{(Ux, x) : x \in \ell_1\}.$$

Also define the tail operator

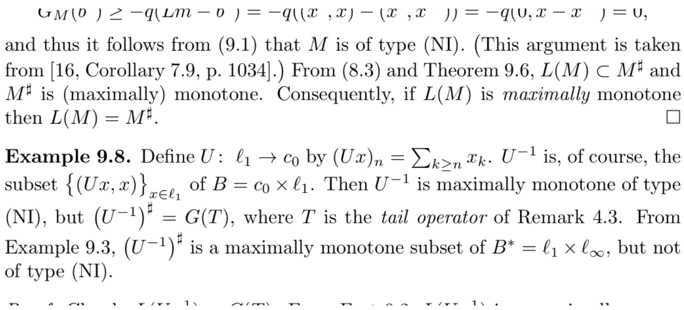
$$T : \ell_1 \longrightarrow \ell_\infty, \qquad (Tx)_n = \sum_{k\ge n} x_k.$$

**Proposition 1.** *The multifunction $S$ is closed, monotone, and quasidense, but its Fitzpatrick extension $S^{\mathrm{F}}$ is not quasidense.*

*Proof.* Simons proves that $G(S) = U^{-1}$ is maximally monotone of type (NI), hence quasidense, and that

$$G(S^{\mathrm{F}}) = G(S)^{\sharp} = G(T).$$

He separately proves that $T$ is maximally monotone but not of type (NI) [2, Examples 9.3 and 9.8]. Since type (NI) and quasidensity are equivalent here, $S^{\mathrm{F}} = T$ is not quasidense. This is the requested negative answer. □



and thus it follows from (9.1) that $M$ is of type (NI). (This argument is taken from [16, Corollary 7.9, p. 1034].) From (8.3) and Theorem 9.6, $L(M) \subset M^{\sharp}$ and $M^{\sharp}$ is (maximally) monotone. Consequently, if $L(M)$ is *maximally* monotone then $L(M) = M^{\sharp}$. □

**Example 9.8.** Define $U\colon\ \ell_1 \to c_0$ by $(Ux)_n = \sum_{k\ge n} x_k$. $U^{-1}$ is, of course, the subset $\big\{(Ux, x)\big\}_{x\in\ell_1}$ of $B = c_0 \times \ell_1$. Then $U^{-1}$ is maximally monotone of type (NI), but $\big(U^{-1}\big)^{\sharp} = G(T)$, where $T$ is the *tail operator* of Remark 4.3. From Example 9.3, $\big(U^{-1}\big)^{\sharp}$ is a maximally monotone subset of $B^* = \ell_1 \times \ell_\infty$, but not of type (NI).

Figure 2: The explicit counterexample in arXiv:2407.00479, PDF page 13.

For completeness, the later paper certifies failure of type (NI) directly. Taking $y^* = (1, 1, \ldots) \in \ell_\infty$ and a Banach limit $y^{**} \in \ell_\infty^*$, it obtains, for every $x \in \ell_1$ and $\sigma = \sum_n x_n$,

$$\langle Tx - y^*, \widehat{x} - y^{**}\rangle \ge \tfrac{1}{2}\sigma^2 - \sigma + 1 = \tfrac{1}{2}(\sigma - 1)^2 + \tfrac{1}{2} \ge \tfrac{1}{2}.$$

Thus the negative-infimum condition fails by a strict margin.

# 3 Assessment

This is an exact, later primary-source answer to a concrete question in the target paper. It is not merely a related result: the later paper explicitly states that the construction gives a negative answer to the earlier problem, and its $M^{\sharp}$ is the same graph as the source's $S^{\mathrm{F}}$. Accordingly, the target should be archived as literature-already-answered rather than promoted as a new proof or counterexample.

The other broad remarks in arXiv:1612.02500—whether all maximally monotone operators are type (FPV)/strongly maximal and whether certain proofs can avoid the bidual—are distinct, long-standing programmatic questions and are not claimed resolved here.

# References


[1] Stephen Simons, *Quasidense monotone multifunctions*, arXiv:1612.02500v4, 2018. `https://arxiv.org/abs/1612.02500`.

[2] Stephen Simons, *Faces of quasidensity*, arXiv:2407.00479, 2024. `https://arxiv.org/abs/2407.00479`.